Bimetric Edges on a Shared Skeleton: Grounding an Eisenstein Dark Sector in Regge Calculus

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Abstract

We begin from the speculative dark-sector dispersion relation

$$E^2 = (pc)^2 + (mc^2)(pc) + (mc^2)^2,$$

whose mixed term is precisely that of the Eisenstein quadratic form $X^2 + XY + Y^2$. This note asks what follows if that observation is taken geometrically. On a single Regge simplicial complex we assign baryonic edge lengths using the standard quadratic form and dark edge lengths using the Eisenstein form. The same off-diagonal half that produces the dispersion cross-term then changes simplex angles and Regge deficit angles. If the deformation is allowed to vary from simplex to simplex, neighbouring developments can also fail to agree in length across a shared face, producing a discrete tear field. With the natural symmetric connection this is face-concentrated nonmetricity; a torsion interpretation is available only if a globally consistent metric-compatible discrete connection can be constructed whose translational holonomy reproduces the tears. No such connection is proved here. The construction is therefore deliberately kinematic: the dispersion relation is an ansatz, no action is supplied for the deformation parameter, and no dark-matter phenomenology is fitted. The novelty claimed is narrower: the Eisenstein dispersion relation and a bimetric Regge geometry can be read as two uses of the same quadratic form, with a concrete discrete defect structure when the two length rules vary across the complex.

1 The Eisenstein dispersion conjecture

The starting point is the proposed dark-sector dispersion relation

$$E^2 = (pc)^2 + (mc^2)(pc) + (mc^2)^2. \tag{1}$$

Its distinctive feature is the mixed mass–momentum term. Write $X = pc$ and $Y = mc^2$. Then $X^2 + XY + Y^2$ is the Eisenstein quadratic form. Equivalently, with $v = (X, Y)^\mathsf{T}$ and

$$G_\mathrm{d} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix},$$

the conjecture is $E_\mathrm{d}^2 = v^\mathsf{T} G_\mathrm{d} v$. For comparison the standard relation is obtained from $G_\mathrm{b} = I$. Write the kinematic state as the vector $v = (X, Y)^\mathsf{T}$ on the energy plane, $X = pc$, $Y = mc^2$; then both dispersion relations are the single statement $E^2 = v^\mathsf{T} G\, v$, read with each sector's Gram matrix:

$$E_\mathrm{b}^2 \;=\; v^\mathsf{T} G_\mathrm{b}\, v \;=\; X^2 + Y^2, \qquad E_\mathrm{d}^2 \;=\; v^\mathsf{T} G_\mathrm{d}\, v \;=\; X^2 + XY + Y^2. \tag{2}$$

Introducing $G(\varepsilon) = \left(\begin{smallmatrix} 1 & \varepsilon/2 \\ \varepsilon/2 & 1 \end{smallmatrix}\right)$ gives $v^\mathsf{T} G(\varepsilon) v = X^2 + \varepsilon XY + Y^2$. The coefficient of the mixed term is therefore the same off-diagonal half that characterises the Eisenstein form. This is an algebraic observation, not a derivation of the dispersion law from dynamics.

The ansatz retains the standard rest and massless limits. At rest, $X = 0$ and $E_\mathrm{d} = mc^2$; for a massless particle, $Y = 0$ and $E_\mathrm{d} = pc$. The mixed term is nonzero only when both mass and momentum are nonzero.

2 From dispersion to a bimetric skeleton

The dispersion ansatz above suggests a geometric question: can the Eisenstein quadratic form be used not only on the energy plane but also as a second rule for measuring edges on a shared discrete spacetime? The proposal explored here is deliberately minimal. There is one simplicial complex and two sectoral length assignments. The baryonic sector uses $G_b = I$; the dark sector uses G_d. No second dynamical spin–2 field is introduced, and no claim is made here that this kinematic construction explains observed dark-matter phenomenology.

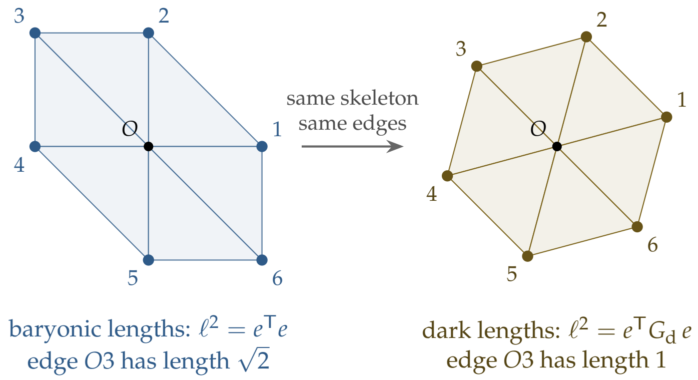


Figure 1: The defining picture of the paper. Left and right show the *same* simplicial complex: identical vertices, identical edges, identical gluing. What differs is the rule each sector uses to convert an edge into a length, the baryonic identity Gram matrix on the left, the Eisenstein Gram matrix on the right (the right panel is drawn in the chart where dark lengths appear as true lengths). Some edges keep their length across the two readings ($O1$, $O2$), others do not ($O3$: $\sqrt{2}$ baryonic, 1 dark), and that selective disagreement is the source of every effect in the paper. There are not two lattices; there is one skeleton and two rulers.

The figure is the essential bimetric statement: the combinatorics are unchanged while the metric data differ. For an edge vector $e = (x, y)^\mathsf{T}$,

$$\ell_b^2 = e^\mathsf{T} G_b e = x^2 + y^2, \qquad \ell_d^2 = e^\mathsf{T} G_d e = x^2 + xy + y^2.$$

The one-parameter interpolation

$$G(\varepsilon) = \begin{pmatrix} 1 & \varepsilon/2 \\ \varepsilon/2 & 1 \end{pmatrix}, \qquad 0 \le \varepsilon \le 1,$$

will later allow neighbouring simplices to carry different amounts of the Eisenstein deformation. G_d is positive definite, so nothing here requires a Finsler norm: the departure is the coexistence of two quadratic forms on the same edge data.

The question is now concrete. Regge calculus constructs curvature entirely from edge lengths. If the Eisenstein form supplies a second set of lengths on the same complex, it must also supply a second set of simplex angles and deficit angles. That is the construction of the next section.

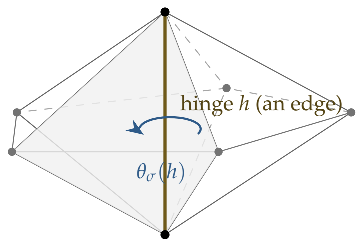
3 The Eisenstein–Regge construction

Regge calculus replaces a smooth metric by edge-length data on a simplicial complex. This makes it a natural setting in which to test a second quadratic length rule, because changing the edge lengths changes the simplex angles and hence the discrete curvature.

Regge's discretisation of general relativity replaces the smooth manifold by a simplicial complex and the metric tensor by the assignment of a length to every edge. Each simplex is internally flat; curvature cannot hide inside a simplex and is instead concentrated on the hinges, the codimension-two faces (triangles in four dimensions, edges in three, vertices in two), where the surrounding simplices fail to fill the full angle (Figure 2). The failure is the deficit angle

$$\epsilon_h \;=\; 2\pi - \sum_{\sigma \supset h} \theta_\sigma(h), \tag{3}$$

where $\theta_\sigma(h)$ is the dihedral angle of simplex σ at hinge h, and the Regge action is the hinge-weighted sum $S = \sum_h A_h\, \epsilon_h$, with A_h the hinge volume.



five tetrahedra sharing one edge; each contributes a dihedral angle at the hinge, and curvature is the failure of those angles to sum to 2π

Figure 2: The hinge in three dimensions. In 3D Regge calculus a hinge is an *edge* (gold), and the simplices meeting around it are tetrahedra, five shown here fanned about the shared edge AB. Each tetrahedron is internally flat and contributes one dihedral angle $\theta_\sigma(h)$ measured around the hinge; the deficit $\epsilon_h = 2\pi - \sum_\sigma \theta_\sigma(h)$ is where all the curvature lives. The two-dimensional pictures elsewhere in this paper are the slice of this figure perpendicular to the gold edge, with the hinge appearing as a single vertex.

The decisive structural fact is that every quantity in this action, the dihedral angles and the hinge volumes alike, is computed from edge lengths through the law of cosines, and the law of cosines is polarisation: recovering the angle between two edges from three lengths presupposes that lengths come from a bilinear form. This is where the second quadratic form enters the discrete theory. The operation used repeatedly below, *developing* the complex, unrolling the simplices around a hinge into a single flat chart, is shown as a sequence in Figure 3; it is the most intuitive picture in Regge calculus and the one on which every computation in §4 runs.

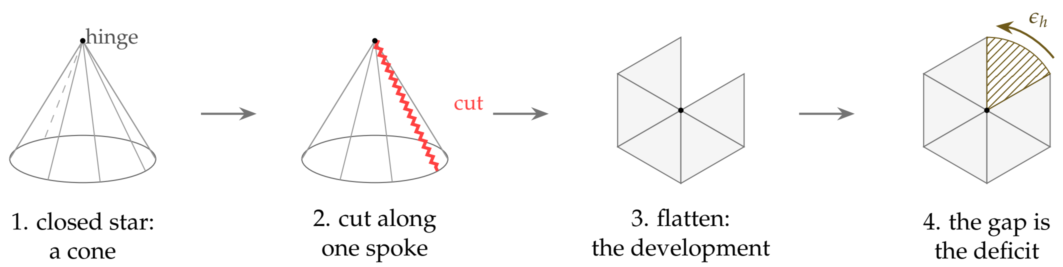
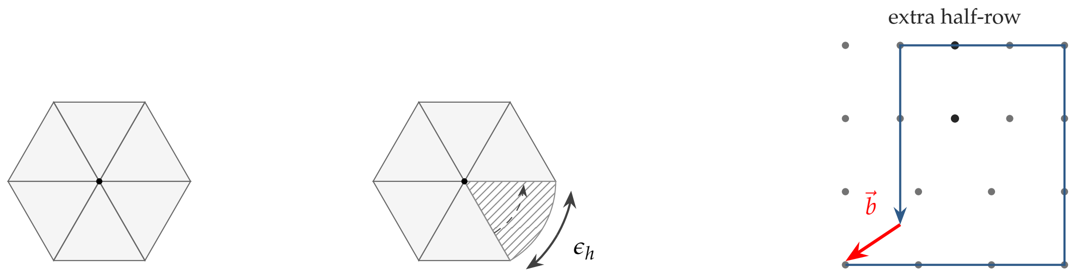


Figure 3: Development, the operation behind every computation in this paper. A closed vertex star is intrinsically a cone (1); cut it along one spoke (2) and unroll the triangles, each internally flat, into a single plane (3); if the star was genuinely curved the unrolled fan fails to close, and the gap (4) is the deficit angle, the curvature the cone was carrying at its hinge. Everything in §4 is this operation performed with two different rulers: when the triangles carry unequal deformations, the fan can fail to close in length as well as in angle, and the length failure is the face tear.

Cursory: deficit angle as curvature

Cut a paper disc, remove a wedge of angle ϵ, and tape the cut edges together; the result is a cone, flat everywhere except at the apex, where all the curvature of the removed wedge now lives. Parallel transport a vector around the apex and it returns rotated by exactly ϵ. Regge curvature applies this construction at every hinge: every hinge is a potential cone apex, and the deficit angle is the holonomy of a loop around it.



(a) flat: six 60° wedges, sum 360°

(b) disclination: five wedges, glue the cut, deficit $\epsilon_h = 60°$

(c) dislocation: equal-step circuit fails to close by $\vec{b}$ (torsion)

Figure 4: Curvature and torsion as lattice defects in standard Regge calculus. (a) Six 60° triangles close the vertex star exactly: flat. (b) Remove one and glue the cut edges (dashed identification): the star becomes a cone, all curvature concentrated at the hinge as the deficit angle ϵ_h, a disclination. (c) The other defect: around a dislocation, a circuit of equal lattice steps fails to close, and the failure $\vec{b}$ is the Burgers vector, the discrete signature of torsion; in three dimensions $\vec{b}$ may point along the hinge itself, the screw dislocation, which is where contorsion proper lives.

3.1 The bimetric assignment

Give every edge e of the complex a vector $\vec{e}$ in the frame of each simplex containing it, and assign it two squared lengths,

$$\ell_{\mathrm{b}}^2(e) \;=\; \vec{e}^{\,\mathsf{T}}\, G_{\mathrm{b}}\, \vec{e}, \qquad \ell_{\mathrm{d}}^2(e) \;=\; \vec{e}^{\,\mathsf{T}}\, G_{\mathrm{d}}\, \vec{e} \;=\; \ell_{\mathrm{b}}^2(e) + \varepsilon\,(e_x e_y), \tag{4}$$

where the final expression, written in two dimensions for clarity, displays the dark deformation as a cross-term between edge components, the simplicial shadow of the kinematic cross-term of equation (1). Each length assignment feeds the Regge machinery independently: baryonic dihedral angles, baryonic deficits, and a baryonic action from ℓ_{b}; dark dihedral angles, dark deficits, and a dark action from ℓ_{d}. The skeleton is shared while the two length assignments generally produce different deficit angles. Under a sector-specific geodesic prescription this would permit distinct effective free-fall structures, but no matter coupling or field equations are supplied here, so that phenomenological interpretation remains outside the present construction.

Remark 1. *If edge vectors are restricted to the Eisenstein lattice itself, the dark squared lengths take values in the Loeschian numbers $a^2 + ab + b^2 \in \{1, 3, 4, 7, 9, 12, 13, \dots\}$, giving a principled, arithmetic version of the discrete Regge model, in which edge lengths are restricted to a finite menu for computability. Here the menu is dictated by the ideal theory of $\mathbb{Z}[\omega]$ rather than by numerical convenience.*

4 Contorsion in the bimetric lattice

The terms need fixing before the mechanism can be stated, and the statement then splits cleanly by regime: the continuum forbids torsion here, and the lattice supplies it.

4.1 Connection and metric: two objects, three failure modes

A *metric* g answers static questions: how long is this vector, what angle do these two make, here, now. A *connection* $\Gamma^{\lambda}{}_{\mu\nu}$ answers a different question: which vector at a neighbouring point counts as *the same* vector as this one, that is, how to transport. Nothing in either definition mentions the other, and a priori a theory carries both as separate ontological commitments. They can be made to talk: the connection is *metric-compatible* when transport preserves the metric's verdicts, $\nabla_{\lambda} g_{\mu\nu} = 0$, lengths and angles unchanged along every journey; and it is *symmetric* when $\Gamma^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\nu\mu}$. The fundamental theorem of Riemannian geometry says that demanding both at once leaves exactly one connection, the Levi-Civita connection $\mathring{\Gamma}[g]$, computable from g alone. General relativity imposes both demands, which is why its connection appears to be a derivative of its metric; metric-affine gravity [23] declines to impose them, restoring the connection to independent life, and the failures of the two demands are then measured by two tensors of entirely different character:

$$T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu} \qquad \text{(torsion)}, \qquad\qquad Q_{\lambda\mu\nu} = \nabla_{\lambda} g_{\mu\nu} \qquad \text{(nonmetricity)}. \tag{5}$$

Torsion is a property of the connection alone, its antisymmetry, and means that infinitesimal parallelograms fail to close. Nonmetricity is a property of the *pair*, connection against metric, has nothing to do with antisymmetry, and means that transported rulers drift. Any connection then splits relative to a chosen metric as

$$\Gamma^{\lambda}{}_{\mu\nu} = \mathring{\Gamma}^{\lambda}{}_{\mu\nu}[g] + K^{\lambda}{}_{\mu\nu} + L^{\lambda}{}_{\mu\nu}, \tag{6}$$

with the contorsion K built algebraically from T and the disformation L built from Q: two correction terms, each repackaging its failure tensor into connection-shaped form.

On the Regge lattice the metric is the edge-length assignment and the connection is the set of face gluing maps of the development; a metric-compatible gluing is a length-preserving one. The baryonic sector uses its Levi-Civita pair throughout, edge lengths from G_{b} and length-preserving rotational gluings, so its torsion and nonmetricity both vanish and its entire geometry is curvature. The dark sector, taken alone, does the same with G_{d}. All the new structure lives *between* them: transport dark lengths by baryonic gluings, or glue across a face where ε_{σ} jumps, and the pair fails compatibility, which is nonmetricity, hence disformation in (6); whether any of it can instead be represented as torsion and contorsion is the channel question the following subsections resolve.

4.2 The continuum verdict: pure disformation

Let each sector use the Levi-Civita connection of its own metric, written with bracketed labels to keep sector tags away from tensor indices: $\mathring{\Gamma}[g_{\mathrm{b}}]$ and $\mathring{\Gamma}[g_{\mathrm{d}}]$, with $g_{\mathrm{d}} = G(\varepsilon(x))$. The difference of two connections is always a tensor, and here it is a disformation:

$$L^{\lambda}{}_{\mu\nu} = \mathring{\Gamma}^{\lambda}{}_{\mu\nu}[g_{\mathrm{d}}] - \mathring{\Gamma}^{\lambda}{}_{\mu\nu}[g_{\mathrm{b}}], \tag{7}$$

symmetric in $\mu\nu$ because both parents are, so no antisymmetric part exists to represent as torsion. Its source is the nonmetricity of the pair: transport the dark metric with the baryonic connection and $\nabla^{[\mathrm{b}]}_{\lambda} g_{\mathrm{d}} = (\partial_{\lambda}\varepsilon)\, S$, the gradient of the deformation directed along the fixed traceless shear S.

Proposition 1. *A bimetric structure in which each sector uses the Levi-Civita connection of its own metric generates no torsion. Its entire content beyond the two curvatures is the disformation* (7)*, sourced by the nonmetricity* $(\partial\varepsilon)\,S$*, and it vanishes wherever* ε *is constant.*

Three clarifications are worth making. First, the locus: the nonmetricity belongs to *neither* sector. Inside the baryonic sector, baryonic transport preserves baryonic lengths; inside the dark sector, likewise; the failure appears only when one sector's transport is measured against the other's metric, so it is a strictly relational quantity, a relational geometry. Second, the Weyl comparison, stated exactly: nonmetricity decomposes into a trace part, a common rescaling of all lengths, which is Weyl's original scale gauge [29], and a traceless part that stretches directions unequally; since S is traceless, the present nonmetricity is *purely* of the second kind, shear with no Weyl scaling at all, so no transported dark clock runs fast or slow overall, it is only that the two diagonal directions trade calibration. The gauge-theoretic reading of this shear is developed in §4.4. Third, the corollary: constant ε makes (7) vanish on flat space and the bimetric structure globally invisible, the lone-inner-product remark again; the deformation must vary for anything to happen, and when it does, the continuum home of the proposal is symmetric teleparallel and metric-affine territory, not Einstein–Cartan. Torsion, and with it contorsion, will enter only as an optional rebooking on the lattice, in the following subsection.

4.3 The lattice verdict: tears, and the choice of channel

On the Regge lattice the natural bimetric assignment is not a smooth $\varepsilon(x)$ but a piecewise-constant one: each simplex σ carries its own deformation ε_σ, constant on the flat interior, jumping across the shared face when one passes to a neighbour. Developing the complex, unrolling the simplices around a hinge into a single flat chart, requires gluing each simplex to the next along the shared face, and the gluing can always match the face's direction by a rotation about the hinge, while it can match the face's length only when the two simplices' norms agree on it. Where they disagree, the two images of the far vertex split, and the gap

$$b_k \;=\; (\text{image of vertex } k \text{ from simplex } \sigma_k) \;-\; (\text{image from simplex } \sigma_{k-1}) \tag{8}$$

is the raw observable of the construction: a gluing failure, prior to any geometric interpretation.

The interpretation is a choice, and this is the point at which the apparent contradiction with the continuum proposition dissolves. The lattice data, edge vectors and two length rules, does not by itself say whether a tear is torsion or nonmetricity; that attribution belongs to the discrete connection one builds on top of the data, exactly as the metric-affine decomposition (6) says it must. Two readings are available. Under the *symmetric* reading, the one used in the development above, each simplex transports with its own rule and the tear records the metric itself changing across the face: the same edge carries two lengths, a wall of mismatched lengths in the crystal dictionary, which is nonmetricity concentrated on faces, and its smooth limit is precisely the disformation of the continuum proposition. The lattice and continuum verdicts agree. Under the *metric-compatible* reading, one instead insists that transport preserve lengths, in the manner of Drummond [17] and of Caselle, D'Adda and Magnea [18], and the same gluing failure, now forbidden from the length channel, is forced into the translation channel, where it is a Burgers vector and reads as torsion, the dislocation of [19]. This trade of nonmetricity for torsion at fixed physical content is the geometric trinity of the companion note operating at the level of a single face, and it recasts the open verification of §4.5 in its correct form: not whether the tears *are* torsion, which is a category error, but whether a globally consistent metric-compatible discrete connection exists whose torsion field is the tear field, satisfying the Drummond transformation properties and Bianchi constraints. If it does, the dark sector admits a genuine torsion description; if it does not, the tears are irreducibly nonmetricity and the framework's discrete geometry is symmetric-teleparallel in character throughout.

4.4 The dark gauge: Weyl's idea in the fractures

The channel analysis can be given a gauge-theoretic voice, and doing so makes precise the sense in which the fractures carry Weyl's nonmetricity. The lattice framework has the anatomy of a lattice gauge theory of the metric-affine group: the gauge freedom is the choice of flat chart within each simplex, the local frames of the development; the connection is the set of face transition maps gluing neighbouring frames; and the physical, gauge-invariant content is the holonomy around hinges, whose rotational part is curvature and whose remaining parts are the defect content this section has been dissecting. What the bimetric deformation adds to the ordinary Regge gauge structure is exactly one new feature: the transition maps across faces of unequal ε_σ can no longer be taken orthogonal, because they must relate frames calibrated by different Gram matrices, and their non-orthogonal part is a length change. A connection whose transition maps rescale lengths is precisely what Weyl introduced in 1918 [29]: his generalisation of Riemannian geometry allowed parallel transport to alter magnitudes, with the alteration recorded in a nonmetricity sourced by a gauge potential, and his ambition, premature but structurally prophetic, was to read that potential as electromagnetism. The present framework is a sector-restricted, discrete, and shear-type descendant of that idea. The tears are the flux of the length-change part of the transition maps through the faces, a discrete Weyl nonmetricity; it is shear-type rather than Weyl's original scale-type, since §A's traceless S deforms directions unequally where Weyl's factor rescaled them all alike, but the gauge anatomy, potential on the gluing data, flux on the defects, invariance under reframing, is his. In this voice the two sectors run two gauge structures in unison on one skeleton: the baryonic sector gauges rotations, its curvature the ordinary Regge deficit, while the dark deformation additionally gauges the shear direction, its "field strength" the tear field, and the Vierergruppe of equation (10) is the residual global symmetry the two gaugings share.

Cursory: teleparallelism, and why it keeps appearing

General relativity's field can be carried by any one of three geometric channels: curvature (the standard formulation), torsion (teleparallel gravity, TEGR, built on the Weitzenböck connection, whose curvature vanishes identically and whose frames are parallel at a distance), or nonmetricity (symmetric teleparallel gravity, STEGR, where both curvature and torsion vanish and lengths drift instead). The three actions differ by boundary terms and yield identical field equations [24]; teleparallel Regge calculus exists [22], so the trade is available on the lattice as well. The framework of this paper keeps encountering teleparallelism because its natural defects are exactly the teleparallel currencies: force the tears into the translation channel and the dark sector is TEGR-like, torsion carrying its gravity; leave them in the length channel, the reading the continuum limit prefers, and it is STEGR-like, nonmetricity carrying it. The companion note distributes these choices across the sectors deliberately; the point here is only that the choice exists, that it is the same choice Weyl's successors formalised, and that nothing in the lattice data forces it.

The tear field is arithmetically selective

The tear at a face is the difference of the two sectors' lengths for the spoke crossing it, so it vanishes identically on spokes whose Eisenstein and Euclidean norms coincide, and is nonzero precisely on the norm-sensitive spokes. Among the six Eisenstein units, the four spokes $\pm(1,0)$, $\pm(0,1)$ have squared length 1 in both sectors, while $\pm(-1,1)$ have squared length 2 baryonic and 1 dark. A dark domain wall therefore carries defect content or does not according to the number theory of its boundary edges, and the selection rule is channel-independent: whichever reading one adopts, nonmetricity wall or dislocation wall, the arithmetic decides where the defect can live. The lattice admits silent walls, invisible in the defect content of the complex, alongside loud ones.

4.5 A worked example, and the claim that actually needs proving

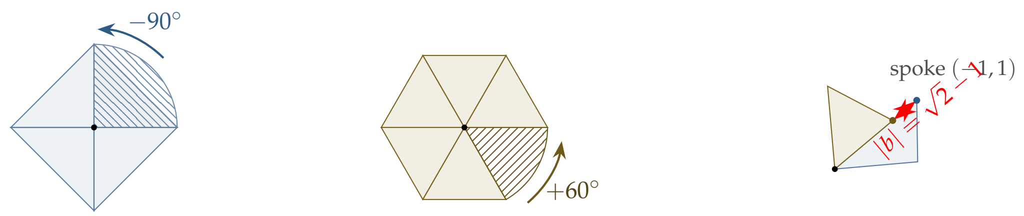
The mechanism is best seen on the smallest complex that exhibits it, the six-triangle vertex star with per-triangle ε_σ, and it is worth being candid about the status of what follows: these are illustrations of the construction, not findings. The central fact is one line. The tear at a face is, by construction, the difference between the two triangles' lengths for the spoke crossing it,

$$|b| \;=\; \left|\, \ell_{\varepsilon_\sigma}(e) - \ell_{\varepsilon_{\sigma'}}(e) \,\right|, \tag{9}$$

so every property of the tear, its magnitude, its scaling, and its vanishing, is elementary once the norms are written down. For the spoke $(-1, 1)$ between a dark and a baryonic triangle this gives $\sqrt{2} - 1$; the small-jump behaviour is the first-order expansion of $\sqrt{2 - \varepsilon}$; and the tear vanishes identically on spokes with $e_x e_y = 0$, where the cross-term of the deformation has nothing to act on, which is all the "arithmetic selectivity" of the previous subsection amounts to at the level of a single face: the deformation is a pure cross-term, so its effects live exactly where edge components mix. None of this is deep, and the worked example earns its place only as a concreteness check that the machinery does what its definitions say.

The claim that carries weight, and that equation (9) does not by itself establish, is whether the tears admit a torsion description at all: as §4 establishes, the tear is channel-ambiguous, reading as nonmetricity under symmetric development and as torsion only if a consistent metric-compatible discrete connection exists whose torsion field it is. Two things must hold for that connection to exist and the torsion reading to be more than a borrowed word. First, the tear must be invariant under the freedoms of the construction: the choice of development chart, the ordering of the gluing, and the frame identifications across faces, since a quantity that can be gauged away by re-developing the same complex is bookkeeping, not geometry. The magnitude (9) is manifestly independent of chart and ordering, being a difference of two lengths each computed within a single simplex, but the full Burgers *vector*, direction included, transforms under the development freedom, and the invariant content is properly the conjugacy class of the affine holonomy around the hinge, which has not been computed here. Second, the identification must reproduce the established constructions in their domain: Drummond's torsion [17] and the dislocation picture of Schmidt and Kohler [19] carry specific transformation properties and Bianchi-type constraints, and showing that bimetric tears satisfy them, rather than merely resembling them, is the substantive open task. It stands first in the programme below, ahead of any extension to higher dimensions, because the two outcomes are two different geometries, and the names should be exact. If the identification fails, the tears are irreducibly nonmetricity, the relational quantity of §4.2, and the framework is symmetric-teleparallel in character. If it succeeds, the dark sector becomes a discrete Riemann–Cartan geometry, hinge deficits carrying its curvature and face tears its torsion, with the further teleparallel option of trading the deficits into torsion as well, leaving a flat dark sector whose gravity is entirely dislocation content beside a Riemannian, purely curved baryonic one. Einstein–Cartan theory proper is the *dynamical* completion of the Riemann–Cartan case, with spin as the source of torsion, and the name is reserved for it below.

The crossing values, for completeness. The deficit of an n-star of fundamental triangles vanishes at $\varepsilon_n = 2\cos(2\pi/n)$, each an algebraic integer of the real cyclotomic field $\mathbb{Q}(\zeta_n)^+$: $\varepsilon_4 = 0$, $\varepsilon_5 = 1/\varphi$, $\varepsilon_6 = 1$, $\varepsilon_8 = \sqrt{2}$. The endpoints ε_4 and ε_6 being exactly the two sectors restates the design fact that the square and hexagonal closures are the two crystallographic ones; the intermediate values are forced trigonometry of the chosen n and carry no significance.



(a) baryonic, $\varepsilon = 0$: apex $90°$, five wedges overlap, deficit $-90°$

(b) dark, $\varepsilon = 1$: apex $60°$, five wedges leave gap, deficit $+60°$

(c) the tear: sectors agree on the direction, disagree on the length

Figure 5: The bimetric view, in the two sector colours. Panels (a) and (b) develop the *same* combinatorial five-triangle star: the baryonic sector (blue) measures each apex at $\arccos(0) = 90°$, so the wedges overlap and the vertex carries deficit $-90°$; the dark sector (gold) measures the identical star at $\arccos(1/2) = 60°$, leaving a gap of $+60°$. One skeleton, two curvatures of opposite sign; the crossing value for this star, $\varepsilon_5 = 2\cos 72°$, is pentagon trigonometry forced by the choice $n = 5$. Panel (c) shows the torsion mechanism: two triangles of unequal ε_σ glued along the norm-sensitive spoke $(-1, 1)$ agree on its direction, which a rotation can always arrange, and disagree on its length, $\sqrt{2}$ against 1, so the far vertex splits by the Burgers vector $|b| = \sqrt{2} - 1$ of equation (8). On norm-agnostic spokes the two colours would meet at the same point and the face would carry no torsion.

4.6 Conditional physical interpretations

If the torsion representation survives the verification of §4.5, several physical interpretations become available. They are consequences of an additional connection choice, not results established by the lattice data alone. Under every reading, the invariant statement is the one already established: within each sector, geometry is metric-compatible and torsion-free; between the sectors sits relational nonmetricity. The channel choice concerns only how the dark sector represents its *internal* defect content, and the options are: dark Riemann–Cartan, deficits as curvature and tears as torsion; or dark teleparallel, deficits traded into torsion as well, a flat dark geometry of pure dislocation content beside a Riemannian baryonic one, which is the distribution the companion note develops. Neither option touches the baryonic sector, and neither removes the inter-sector nonmetricity.

First, then, under the torsion rebooking the dark sector carries a Riemann–Cartan-like structure whose torsion is sourced geometrically, by the granularity of the deformation field, rather than by matter: torsion marks where the dark deformation is inhomogeneous, concentrated on domain boundaries and filtered by the arithmetic selection rule, so the dislocation content of a region reads off both the geography of the dark domains and the residue classes of their walls.

Second, the reason to care which channel the dark sector uses, rather than treating the choice as bookkeeping, is fermions. A minimally coupled Dirac field is the one standard piece of matter that feels torsion directly, through the axial part of the contorsion in its spin connection, while remaining blind to the choice of boundary term in the gravitational action; this is what elevates Riemann–Cartan geometry to Einstein–Cartan *theory* when spin sources are present. If dark fermions exist, such a torsion representation could therefore matter through the spin connection. Any observable portal to baryonic fermions would require an explicit interaction Lagrangian, which is not constructed here.

Third, the structure is finitely computable throughout: transition maps are small matrices, tears are vector differences, and the full defect content of a complex, in whichever channel it is booked, is linear algebra over $G(\varepsilon)$.

5 The computational programme

The worked example of §4.5 illustrates the mechanism; the programme ahead must first establish it and then scale it.

1. **The torsion identification.** Compute the affine holonomy around a hinge with mixed ε_σ, extract its conjugacy class, and verify that the translational part transforms as Drummond's discrete torsion and satisfies the corresponding Bianchi-type constraints; this is the first necessary check, and subsequent torsion claims are conditional on it.

2. **Three and four dimensions.** Repeat the development around a hinge (an edge in 3D, a triangle in 4D) with per-simplex ε_σ; the Burgers vector acquires components parallel to the hinge (screw dislocations) alongside the perpendicular tears (edge dislocations) seen in 2D, and the screw component is where genuinely three-dimensional contorsion lives.

3. **Contorsion extraction.** Decompose the face transition maps in the discrete analogue of equation (6), isolating the antisymmetric generator, and map its concentration across a complex with a prescribed ε-landscape.

4. **Crossing values and lattice compatibility.** The zero-deficit deformations $\varepsilon_n = 2\cos(2\pi/n)$ single out the crystallographic cases $n = 4, 6$ as the endpoints of the family; the substantive question is not the numerology of the intermediate values but whether any structure of the framework, the Loeschian edge-length menu, the norm-sensitivity selection rule, degenerates or changes character at non-crystallographic ε, where $G(\varepsilon)$ is the Gram matrix of no lattice at all.

5. **Regge dynamics.** Promote ε_σ to a dynamical variable alongside the edge lengths, vary the doubled Regge action, and determine whether domain walls form spontaneously and whether the torsion-silent walls are energetically preferred, which would make the dark sector's dislocation content sparse in a predictable pattern.

6 Scope and limitations

The construction is kinematic throughout, and the debts are stated once, plainly. There is no action for ε: it is inserted, not derived, so nothing distinguishes one ε-landscape from another, fixes domain-wall structure, or addresses stability, and every claim about where defects concentrate is conditional on a stipulated configuration. And there is no physical derivation of the dispersion relation: equation (1) shows that *if* the dark kinematic plane carries the Eisenstein Gram matrix *then* the cross-term follows by polarisation, but not why it should carry that matrix, which a Lagrangian, an equation of motion, or a representation of the Poincaré algebra would have to supply and none does. A completion therefore needs two things: a dark-sector Lagrangian whose on-shell dispersion relation is (1), and an action for ε_σ, most naturally the doubled Regge action of the programme, whose variation makes the deformation a field.

The positive claim is correspondingly narrow. The Eisenstein quadratic form generates the proposed dispersion cross-term on the energy plane and, when used as a second edge-length rule on a Regge complex, generates a second set of simplex angles and deficits. Allowing the deformation to vary between simplices produces a definite face-tear field whose natural symmetric interpretation is nonmetricity and whose possible torsion interpretation requires the additional verification stated above. These are the results of the present construction; dynamics and phenomenology are deferred.

7 Relation to existing bimetric and discrete-gravity constructions

7.1 Bimetric context

The idea of a second metric is nearly as old as general relativity's maturity. Rosen introduced a flat background metric alongside the dynamical one in 1940 [1], and his later bimetric theory of gravitation [2] survived solar-system tests only to be excluded by binary-pulsar energy loss, an early lesson that a second metric earns its keep or dies by its phenomenology. Isham, Salam and Strathdee's f–g gravity [3] coupled a second tensor field to the hadronic sector specifically, and is the closest historical ancestor of the present proposal's sector-restriction: different matter, different metric. The modern era descends from the massive-graviton problem: the Fierz–Pauli linear theory [4], the Boulware–Deser ghost that haunted its nonlinear completions [5], the de Rham–Gabadadze–Tolley resolution [6, 7], and finally Hassan and Rosen's promotion of the reference metric to a dynamical field, yielding ghost-free bimetric gravity [8], reviewed in [9]. Within that framework the massive spin-2 field itself has been proposed as the dark matter [10], while on the modified-dynamics side both Bekenstein's TeVeS [11] and Milgrom's explicitly bimetric MOND [12] use additional metric structure to do work usually assigned to dark particles. The kinematic face of the proposal, a sector-specific dispersion relation, has its established formalism in the Standard Model Extension [13], which catalogues Lorentz-violating operators including terms bilinear in mass and momentum, and in the quantum-gravity phenomenology of modified dispersion relations [14]; geometrically, the licence to consider non-orthogonal norms at all is the Finsler programme [15], which the present construction undershoots deliberately, since the Eisenstein form remains an inner product.

7.2 Discrete-gravity context and departures

Against this lineage the present construction makes three departures. First, the second metric is not a dynamical spin-2 field: it is algebraically rigid, fixed by the Eisenstein norm form up to the single interpolation parameter ε, so the ghost problematic that shaped the modern literature is sidestepped rather than solved, there being no second kinetic sector to propagate a ghost. The cost of that rigidity is honesty about what is dynamical, nothing beyond the shared edge lengths in the minimal version, the per-simplex ε_σ in the dynamical extension of the closing programme; the algebraic form is rigid, while the amplitude ε remains unspecified until a dynamics is supplied. Second, the deformation has an arithmetic origin rather than an effective-field-theory one: the coefficient of the dispersion cross-term is an order-one number fixed by lattice geometry rather than a small Wilson coefficient awaiting bounds, which is why the proposal must be confined to the dark sector, where the orthogonal relation is untested, and why it is falsifiable there in a way Planck-suppressed modifications are not. Third, the natural home is discrete: where the bimetric literature works on smooth manifolds, the present framework is built on the Regge lattice [16] from the start, and §4 shows the discreteness is load-bearing, since the torsion content of the theory exists only there. The discrete side has its own lineage this paper joins: Drummond's piecewise-constant torsion fields on the simplicial lattice [17], the Poincaré gauge formulation of Regge calculus by Caselle, D'Adda and Magnea [18], the identification of torsion with dislocations and curvature with disclinations [19, 20, 21], teleparallel Regge calculus [22], the metric-affine framework housing curvature, torsion and nonmetricity together [23], and the geometric trinity that makes their equivalence precise [24]. The companion note distributes that trinity across the two sectors.

A Erlangen reading of the bimetric pair

A.1 The Erlangen reading of bimetricity

Klein's Erlangen programme [28] defines a geometry by its transformation group: geometric properties are the invariants of the group's action, and nothing else is geometric. In this language the proposal is one sentence: *the two sectors inhabit one point set but constitute two different Klein geometries*, the baryonic sector governed by the group $O(G_\mathrm{b})$ preserving $x^2 + y^2$, the dark sector by $O(G_\mathrm{d})$ preserving $x^2 + xy + y^2$. Each is an ordinary orthogonal group in its own right, $O(G_\mathrm{d}) = A\,O(2)\,A^{-1}$ with $A = G_\mathrm{d}^{1/2}$, and a lone Klein geometry carries no absolute information, the group-theoretic restatement of the undetectability of a lone inner product. The content of bimetricity is the pair, and Klein locates that content exactly: in the intersection of the two groups, the kinematics the sectors genuinely share.

The intersection is a short computation once the off-diagonal half is put where it belongs. Write $G_\mathrm{d} = I + S$ with $S = \left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$ carrying the entire deformation. A transformation M preserving both forms satisfies $M^\mathsf{T} M = I$ and $M^\mathsf{T}(I + S)M = I + S$, and subtracting the first from the second leaves $M^\mathsf{T} S M = S$, so M must be an orthogonal symmetry of the deformation itself; orthogonality turns this into $MS = SM$, and a matrix commuting with S preserves S's eigenvectors, the diagonals $(1, \pm 1)$ with the distinct eigenvalues $\pm 1/2$. Only four orthogonal maps do so:

$$O(G_\mathrm{b}) \cap O(G_\mathrm{d}) \;=\; \{\pm I,\ \pm\mathrm{swap}\} \;\cong\; \mathbb{Z}_2 \times \mathbb{Z}_2, \tag{10}$$

where swap is $(x, y) \mapsto (y, x)$, a symmetry of both forms because each is symmetric in its variables. This is the Klein four-group: the Erlangen analysis of the bimetric pair terminates, fittingly, in Klein's own Vierergruppe. Every continuous rotation and every other reflection is the private symmetry of one sector and a broken symmetry of the other, so the shared kinematics is rigid, a finite group with no continuous part, and the breaking is organised by S's two eigenaxes, which are precisely the norm-agnostic and norm-sensitive directions on which the defect analysis of §4 turns. The arithmetic selectivity of the tear field is the invariant theory of this broken symmetry.

B ε is not a Brans–Dicke scalar

Once ε is described as a field awaiting an action, the natural comparison is to scalar-tensor gravity, and it is worth blocking that comparison explicitly, since it points in the wrong direction. In Brans–Dicke theory and its descendants, a scalar ϕ enters the action through a nonminimal coupling to curvature, a term ϕR, so that ϕ plays the role of a spacetime-dependent gravitational coupling; the theory remains metric-compatible throughout, nonmetricity is zero, and the well-known transformation to the Einstein frame, $g^*_{\mu\nu} = \phi\, g_{\mu\nu}$, is an isotropic trace-channel rescaling that leaves the physical content unchanged and introduces the scalar's kinetic term only as bookkeeping for the frame change.

The deformation parameter ε does not occupy this role. It does not multiply a curvature scalar, and it is not extracted by any conformal transformation, because, as the shear-versus-trace decomposition of $G(\varepsilon) = I + \varepsilon S$ shows, with S automatically traceless, ε lives entirely in the traceless channel of the Gram matrix, while a Brans–Dicke-type scalar by construction occupies only the trace channel. The two are complementary rather than competing descriptions of the same freedom: nothing prevents a genuine dark-sector Brans–Dicke field $\phi(x)$ from being introduced alongside ε, coupling to whatever plays the role of curvature in the doubled Regge action, with the two fields sourcing orthogonal parts of the dark metric's departure from the baryonic one. Until such a field is introduced, however, this paper contains no gravitational-coupling scalar and no claim about a varying effective Newton constant; ε is closer in spirit to a modulus in the sense of a nonlinear sigma model, a coordinate on the space of possible norms, than to a Brans–Dicke dilaton, and its prospective action, a double-well potential plus a nearest-neighbour gradient term of the kind sketched in the programme discussion, is accordingly a Landau-type free energy for a shape parameter rather than a kinetic term for a gravitational coupling. The nonmetricity this framework produces is correspondingly of shear type, sourced by $\partial\varepsilon$ against the fixed direction S, not the Weyl-vector type a scalar-tensor nonmetricity theory would produce from $\partial \ln \phi$.

C Historical note: the Gram matrix and the metric tensor

The Gram matrix used throughout this paper to encode the dark sector's deformation, and the metric tensor of Riemannian geometry, are the same mathematical object reached by two independent nineteenth-century routes, and the coincidence is worth making explicit since the paper leans on both vocabularies interchangeably. Gauss's first fundamental form [25], the coefficients E, F, G expressing a surface's squared line element as $ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$, is a symmetric matrix of inner products of the coordinate tangent vectors; Riemann's habilitation lecture [26] generalised this to n dimensions as $ds^2 = \sum g_{ij}\,dx^i dx^j$, letting the matrix vary smoothly point to point, which is the metric tensor. Independently, and for entirely different reasons, the actuary and mathematician Jørgen Pedersen Gram, working on least-squares approximation and linear independence in abstract inner-product spaces, introduced the determinant and matrix that bear his name [27], with entries $G_{ij} = \langle v_i, v_j \rangle$ for a set of basis vectors. The differential-geometry tradition and the linear-algebra tradition were answering the same question, how does a vector space measure itself internally, from opposite ends: Riemann posits a bilinear form at each point and asks what lengths and angles it implies, while Gram starts from vectors and reads off the bilinear form they carry. The two names persist by convention rather than by any mathematical distinction, "metric" when the matrix is treated as a field over a manifold, "Gram matrix" when it is treated algebraically as the pairing of a fixed basis, and $G(\varepsilon)$ in this paper is used in the latter, algebraic sense throughout, becoming a metric proper only if and when ε is promoted to a field over the complex, which is precisely the promotion §6 identifies as undone.

References

[1] N. Rosen, General relativity and flat space. II, Phys. Rev. **57** (1940) 150.

[2] N. Rosen, A bi-metric theory of gravitation, Gen. Rel. Grav. **4** (1973) 435.

[3] C. J. Isham, A. Salam and J. Strathdee, F-dominance of gravity, Phys. Rev. D **3** (1971) 867.

[4] M. Fierz and W. Pauli, On relativistic wave equations for particles of arbitrary spin in an electromagnetic field, Proc. Roy. Soc. Lond. A **173** (1939) 211.

[5] D. G. Boulware and S. Deser, Can gravitation have a finite range?, Phys. Rev. D **6** (1972) 3368.

[6] C. de Rham and G. Gabadadze, Generalization of the Fierz–Pauli action, Phys. Rev. D **82** (2010) 044020.

[7] C. de Rham, G. Gabadadze and A. J. Tolley, Resummation of massive gravity, Phys. Rev. Lett. **106** (2011) 231101.

[8] S. F. Hassan and R. A. Rosen, Bimetric gravity from ghost-free massive gravity, JHEP **02** (2012) 126.

[9] A. Schmidt-May and M. von Strauss, Recent developments in bimetric theory, J. Phys. A **49** (2016) 183001.

[10] E. Babichev, L. Marzola, M. Raidal, A. Schmidt-May, F. Urban, H. Veermäe and M. von Strauss, Heavy spin-2 dark matter, JCAP **09** (2016) 016.

[11] J. D. Bekenstein, Relativistic gravitation theory for the modified Newtonian dynamics paradigm, Phys. Rev. D **70** (2004) 083509.

[12] M. Milgrom, Bimetric MOND gravity, Phys. Rev. D **80** (2009) 123536.

[13] D. Colladay and V. A. Kostelecký, Lorentz-violating extension of the standard model, Phys. Rev. D **58** (1998) 116002.

[14] G. Amelino-Camelia, Quantum-spacetime phenomenology, Living Rev. Rel. **16** (2013) 5.

[15] C. Pfeifer and M. N. R. Wohlfarth, Causal structure and electrodynamics on Finsler spacetimes, Phys. Rev. D **84** (2011) 044039.

[16] T. Regge, General relativity without coordinates, Nuovo Cimento **19** (1961) 558.

[17] I. T. Drummond, Regge–Palatini calculus, Nucl. Phys. B **273** (1986) 125.

[18] M. Caselle, A. D'Adda and L. Magnea, Regge calculus as a local theory of the Poincaré group, Phys. Lett. B **232** (1989) 457.

[19] J. Schmidt and C. Kohler, Torsion degrees of freedom in the Regge calculus as dislocations on the simplicial lattice, Gen. Rel. Grav. **33** (2001) 1799.

[20] M. O. Katanaev and I. V. Volovich, Theory of defects in solids and three-dimensional gravity, Ann. Phys. **216** (1992) 1.

[21] H. Kleinert, *Gauge Fields in Condensed Matter*, World Scientific, Singapore (1989), part IV.

[22] J. G. Pereira and T. Vargas, Regge calculus in teleparallel gravity, Class. Quant. Grav. **19** (2002) 4807.

[23] F. W. Hehl, J. D. McCrea, E. W. Mielke and Y. Ne'eman, Metric-affine gauge theory of gravity, Phys. Rep. **258** (1995) 1.

[24] J. Beltrán Jiménez, L. Heisenberg and T. S. Koivisto, The geometrical trinity of gravity, Universe **5** (2019) 173.

[25] C. F. Gauss, Disquisitiones generales circa superficies curvas, Comm. Soc. Reg. Sci. Gött. Rec. **6** (1827).

[26] B. Riemann, Über die Hypothesen, welche der Geometrie zu Grunde liegen, Habilitationsschrift, Göttingen (1854), published Abh. Königl. Ges. Wiss. Göttingen **13** (1868).

[27] J. P. Gram, Ueber die Entwickelung reeller Functionen in Reihen mittelst der Methode der kleinsten Quadrate, J. Reine Angew. Math. **94** (1883) 41.

[28] F. Klein, Vergleichende Betrachtungen über neuere geometrische Forschungen (Erlanger Programm), Erlangen (1872); Math. Ann. **43** (1893) 63.

[29] H. Weyl, Gravitation und Elektrizität, Sitzungsber. Preuss. Akad. Wiss. Berlin (1918) 465.